

Becker Friedman Institute for Economics (BFI), University of Chicago

The Becker Friedman Institute for Economics (BFI) is the University of Chicago economic hub for data-driven insights that inform policy. The BFI's paper, *Carbon Prices, Forest Conservation and Reforestation in the Brazilian Amazon*, finds that carbon prices of \$10-25/ton could flip the Amazon from a net carbon emitter to a net carbon sink over 30 years. To expand the paper's reach, the BFI asked the University of Chicago's Data Science Clinic to create an interactive dashboard using the paper's dataset and geospatial data, making its findings intuitive and accessible for journalists, researchers, policymakers, and the general public.

The team developed a dashboard that visualizes historical deforestation patterns (1995-2017) across 1,043 sites in the Brazilian Amazon. The interactive map presents key spatial patterns such as the "arc of deforestation" in the southeast (Figure 1a). Users can explore four data layers — agricultural area, productivity (θ), carbon density (γ), and their ratio — and access detailed metrics, historical trend charts, and comparisons to Amazon-wide averages by clicking any site (Figure 1b). Users can also explore 50-year model projections under different carbon price scenarios (Figure 2).

Figure 1a.

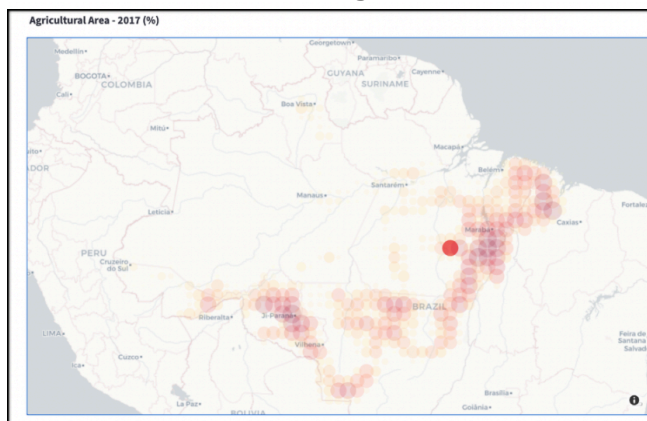


Figure 1b.

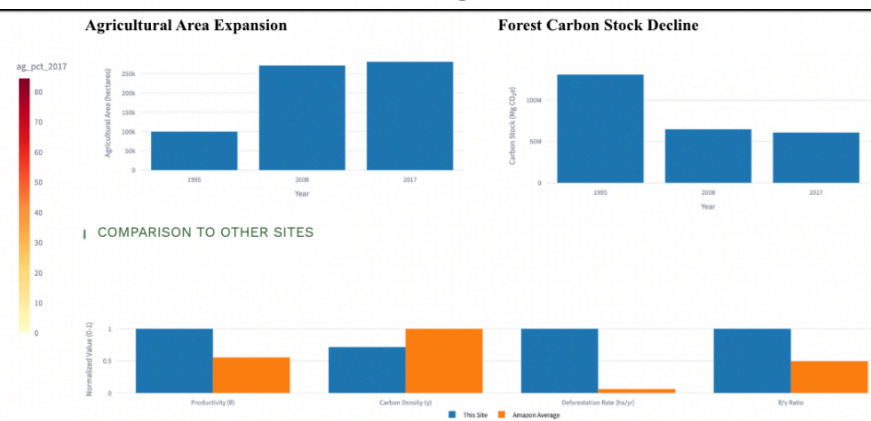
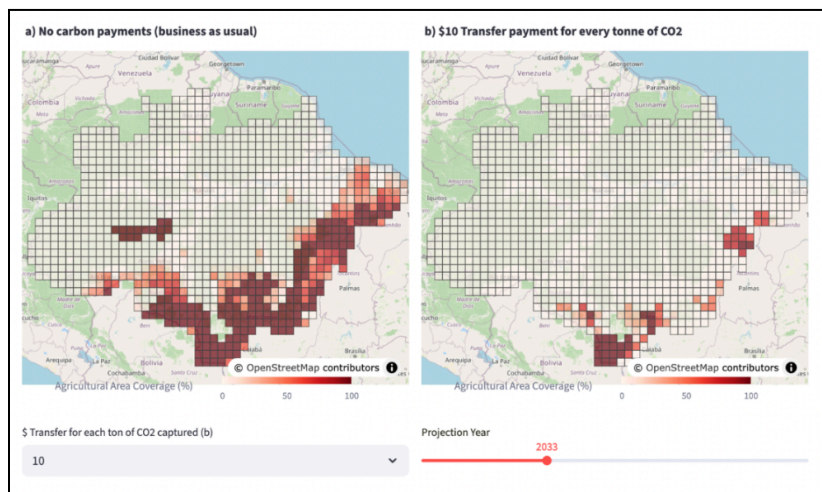


Figure 2.



A central design challenge was translating a mathematically dense economic model into an interface requiring no prior expertise. The team standardized color scales, limited each view to a single variable, and designed the click panel to surface only the most relevant metrics. Rendering geospatial data for over a thousand sites required careful decisions around data preprocessing and tile-based rendering.